Obhoy: A Multi-Line Blockchain Claims-Integrity Protocol for Low-Trust Insurance Markets

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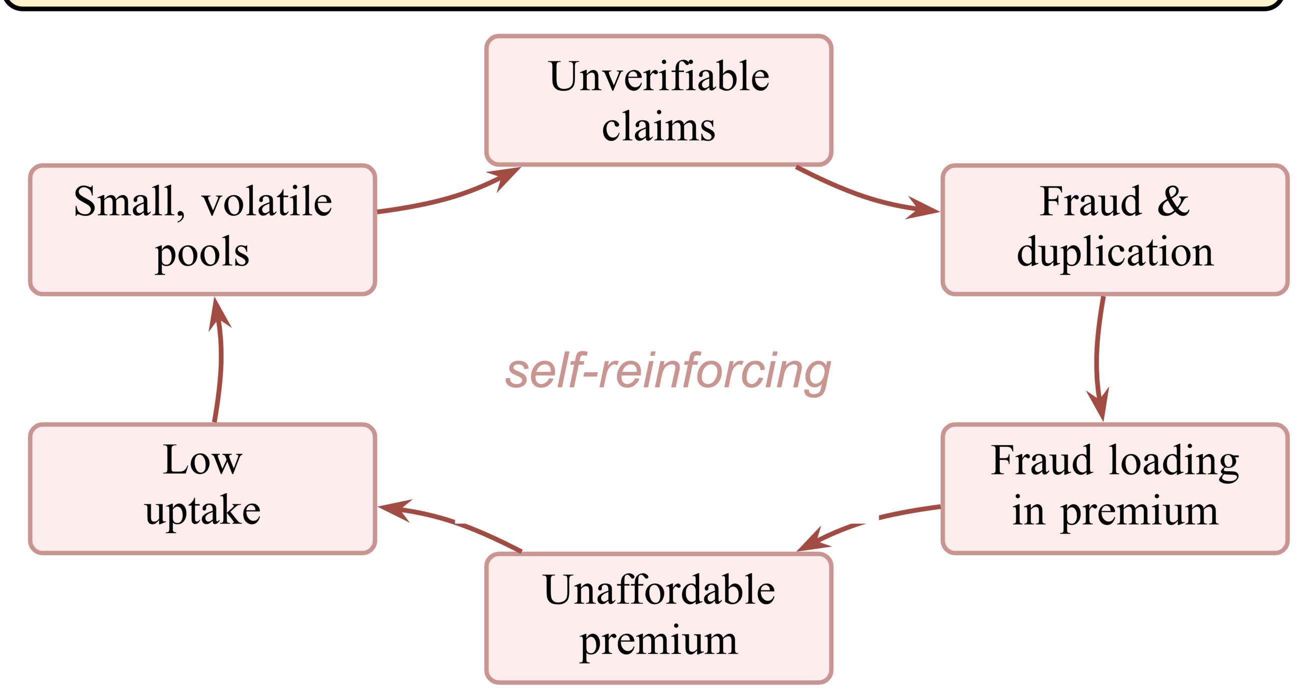
Motivation

➤ **Eliminate fraud margins** by driving verification costs to zero, making coverage viable for low-income households.
➤ **Publish unalterable settlement records** so buyers can independently verify that an insurer actually pays claims.
➤ **Prevent cross-carrier duplicate fraud** through a neutral ledger, protecting families from catastrophic out-of-pocket shocks.
➤ **Provide the missing trust infrastructure** to revive stalled sectors such as : crop, motor, land, health etc. insurances, where a lack of verification prevents markets from forming.
➤ **Replace delayed, self-reported annual filings** with continuous, live auditability for regulators and national health protection programs.

We introduce *Obhoy*, a permissioned claims-integrity ledger that *redefines* the insurable event as a single-use digital asset to *make* insurance verifiable, honest, and affordable.

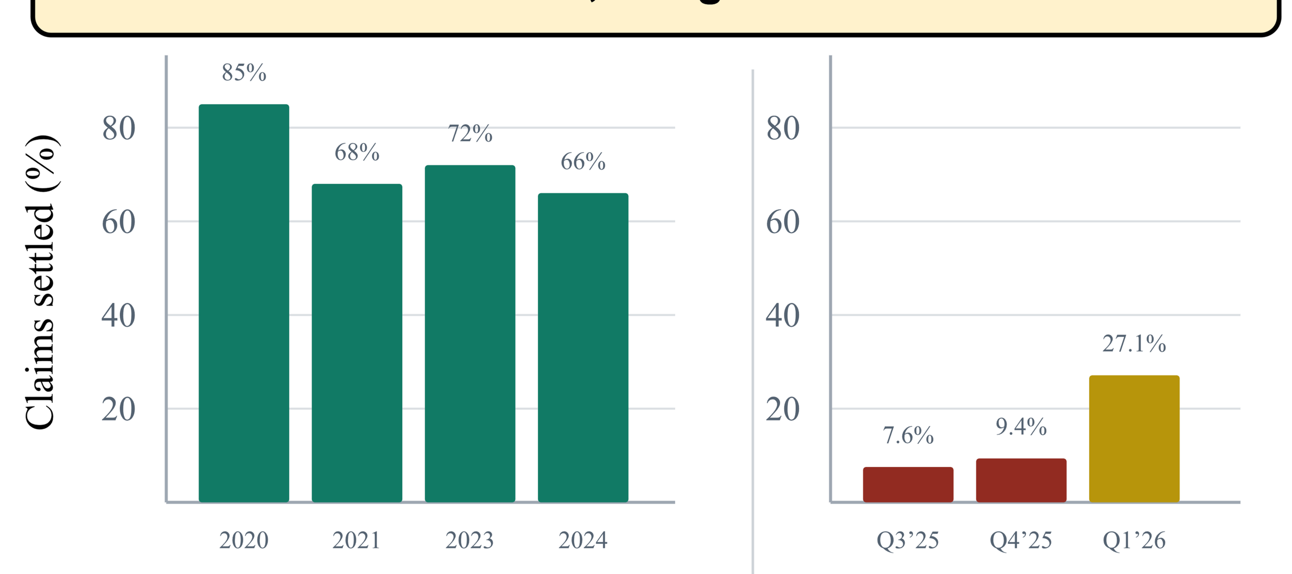
What Problems are we facing?



The Self-Reinforcing failure loop: Each stage amplifies the next, with unaffordable premiums leading to low uptake and small, volatile pools, ultimately returning to unverified claims and perpetuating the cycle.

Obhoy breaks the above loop at its root and turns it into a growth engine by zero-cost verification strips fraud margins from premiums, making policies affordable nd restoring the trust needed to scale adoption

Claims settlement ratio, Bangladesh Insurance Sector



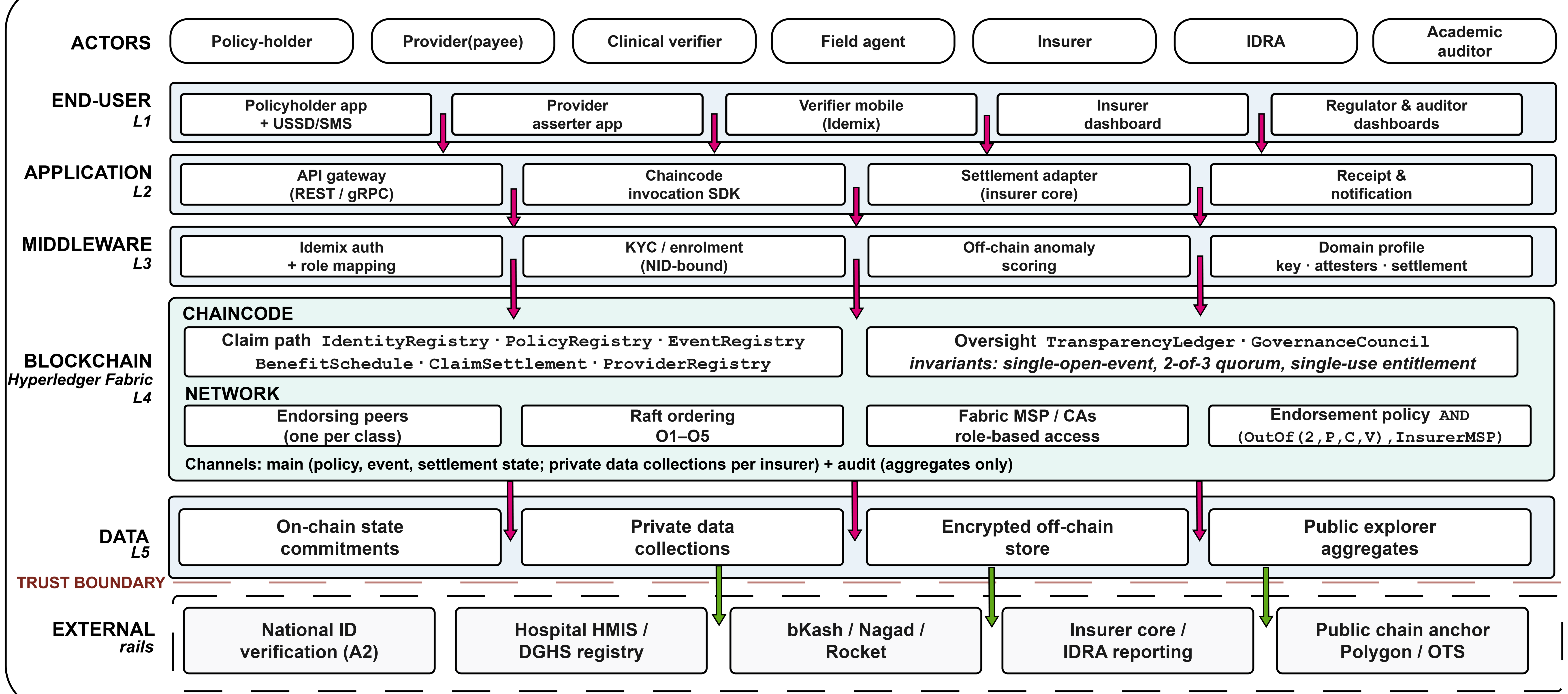
Life Insurance

Annual settlement rates maintain an uneasy baseline between 66% and 85% from 2020 to 2024

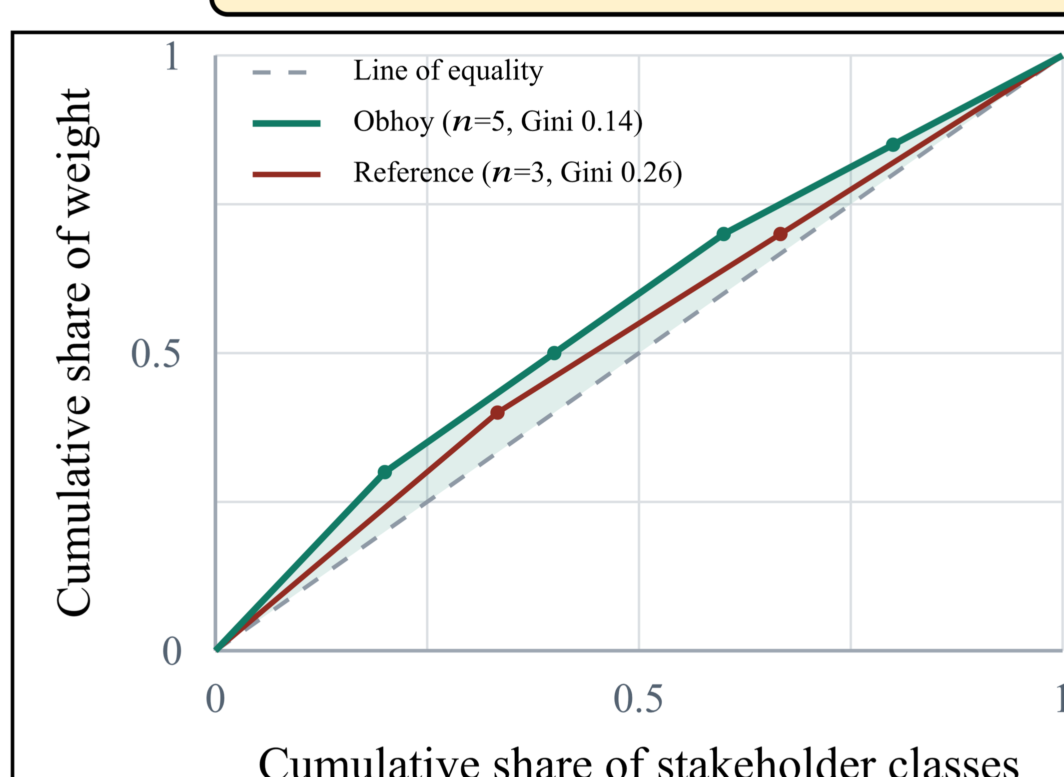
General & Industry

A catastrophic systemic collapse, with general insurers settling fewer than 1 in 10 filed claims by Q4 2025.

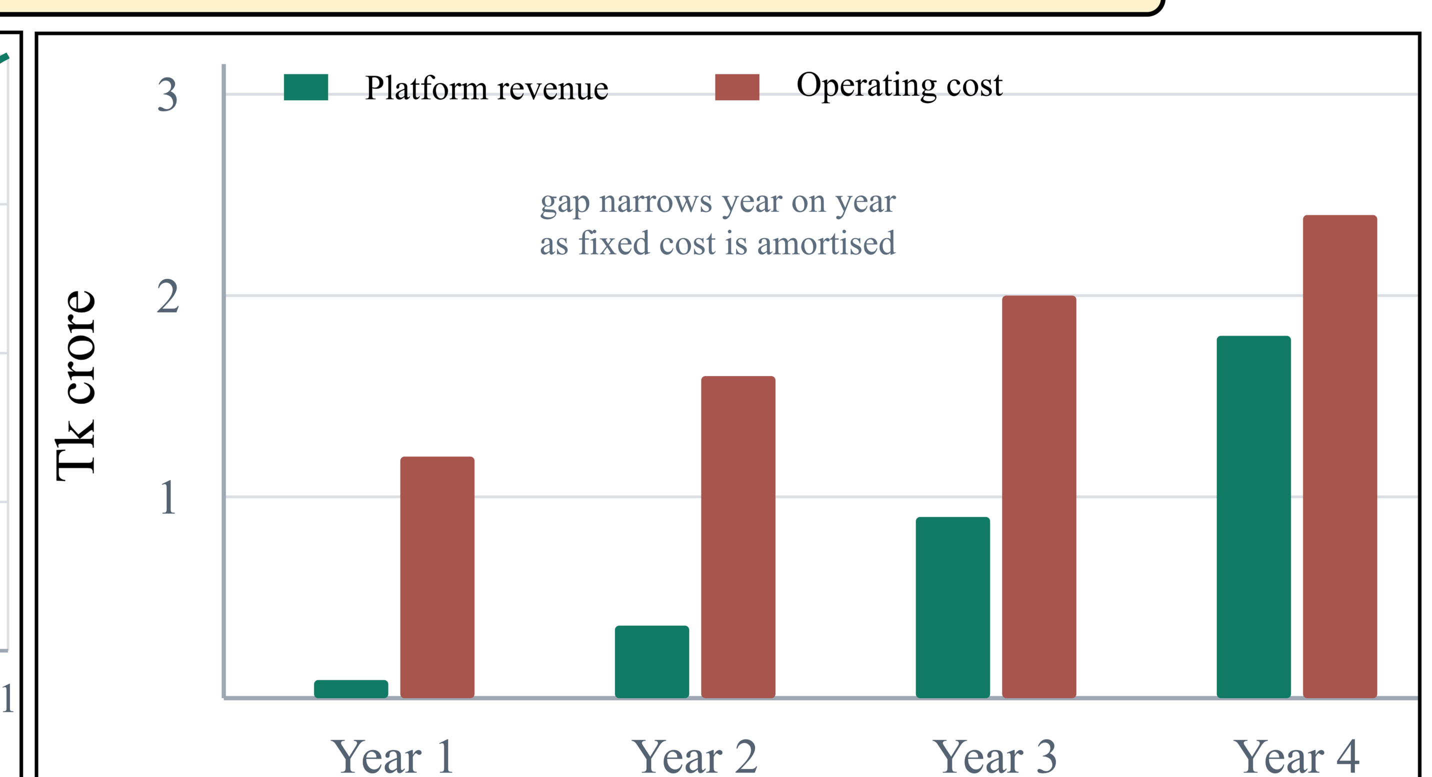
Architecture and Workflow



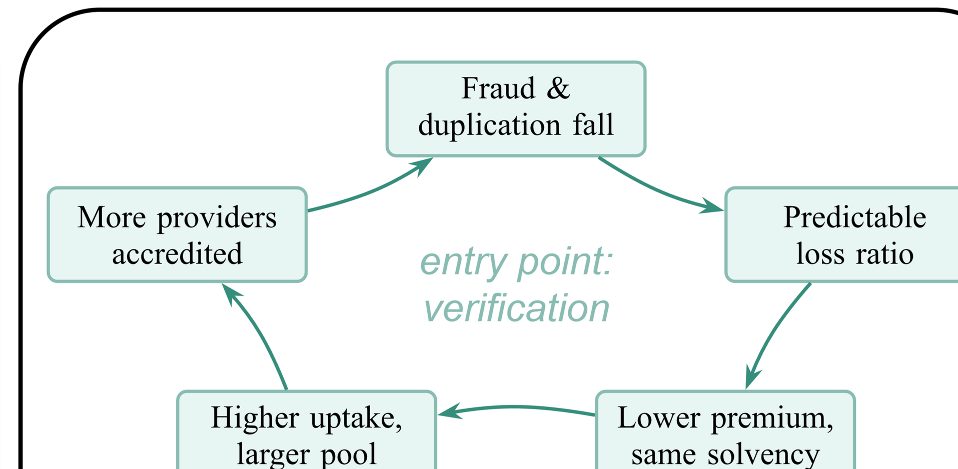
Decentralized Control and Financial Self-Sufficiency



Governance Equality: Lorenz curve of voting distribution across five stakeholder classes ($G = 0.14$, NC = 3$), proving mathematically that notwo institutional classes can collude to control the network.



Financial Sustainability:Illustrative platform revenue versus operating costs (Years 1–4), demonstrating how fixed infrastructure costs amortize toward operational self-sufficiency as scale increases.



The Virtuous Cycle of Verification: Obhoy enters the market through verifiable claims, driving down fraud to lower premiums, which in turn scales adoption and attracts more accredited providers.

Obhoy, routes every claim through a **five-layer execution pipeline** that mathematically forces opposing stakeholders to **verify** an event's uniqueness on-chain before triggering off-ledger mobile payouts and anchoring the final record to a public blockchain.

Implementation

Our code is publicly available!

Conclusion

➤ Where buyers cannot trust an insurer's promise and insurers cannot trust a claimed loss, ***Obhoy's*** shared, multi-party is met with a reliable safety net, not a financial freefall.
➤ Beyond the ledgers and loss ratios, ***Obhoy*** exists for one deeply human reason: to ensure that a family's hardest moment ledger remains the only route to rebuilding that trust and protecting vulnerable households from financial ruin.

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Github: https://github.com/farhanishraq17/Obhoy